

SEMINAR COURSE · BEGINNER'S GUIDE

How to Write a Review Paper

A Step-by-Step Guide for Absolute Beginners

Covers: Understanding · Topic Selection · Literature Search · Structure · Writing · Citation · Submission

What is a Review Paper?

Understanding Before You Start

A review paper is a piece of academic writing that collects, reads, organises, and summarises existing research on a specific topic — without conducting new experiments. Your job is to be a reporter of what others have found.



Narrative Review

Summarises a topic broadly. Most common for beginner students. You read papers and explain what each one found in a logical flow.



Systematic Review

Follows a strict, documented process to search and select papers. Every step is recorded so others can repeat it.



Meta-Analysis

Combines data from multiple studies using statistics to reach one unified conclusion. Requires statistical knowledge.



Scoping Review

Maps out a broad topic to see how much research exists. Good for identifying gaps and future directions.

✦ *For this seminar, you will write a Narrative Review — the simplest and most suitable type for undergraduates.*

Review Paper vs Research Paper

Know the Difference Before You Begin

Aspect	Research Paper	Review Paper
Purpose	Create NEW knowledge through experiments	Summarise EXISTING knowledge from literature
Data	Collected by YOU (surveys, experiments)	Collected by OTHERS (you just read it)
Methodology Section	Required — describes your experiment	Optional/brief — describes your search strategy
Results Section	YOUR data and findings	Summary of what other papers found
Time Required	Months to years	Weeks to months — more manageable!
Skill Level	Advanced — requires lab/field work	Beginner-friendly — requires reading & writing
Example Output	"We tested X and found Y"	"Studies show that X leads to Y"

The 9-Step Roadmap to Writing Your Review Paper

Follow These Steps in Order — Every Time

STEP 1

Choose a Topic

STEP 2

Form Research Questions

STEP 3

Search the Literature

STEP 4

Select Papers (5–8)

STEP 5

Read & Take Notes

STEP 6

Organise by Theme

STEP 7

Write the Paper

STEP 8

Format & Cite

STEP 9

Revise & Submit

Step 1: Choose Your Topic

The Most Important Decision You Will Make

Pick what interests you

You will spend weeks reading about this. If you are bored by the topic, the paper will show it. Choose something you are genuinely curious about.

Be specific — not too broad

'Artificial Intelligence' is a whole field. 'Using CNNs to detect plant diseases' is a paper topic. The narrower, the better.

Check literature availability

Search Google Scholar first. If you cannot find at least 8–10 papers on the topic, it is too narrow. Pick something else.

Stay within your knowledge

Pick a topic you have studied in class or in your project. If you have to learn 5 new fields to understand the papers, it is too hard.

Look for a gap or debate

The best topics are ones where papers disagree or where something is missing. That gap becomes your 'contribution'.

Get your supervisor's approval

In academic courses, confirm your topic before searching. Changing topics after reading 10 papers wastes your time.

 Example good topic for CSE students: "A Review of Deep Learning Methods for Plant Disease Detection"

Step 2: Form Your Research Questions

Turn Your Topic Into Clear Questions

Before you start reading, write 2–4 questions your review will answer. These questions guide everything: what you search for, what papers you select, and how you write.

Example: Topic = Deep Learning for Plant Disease Detection

RQ1: What deep learning architectures are most commonly used for plant disease detection?

RQ2: What datasets are available and how large are they?

RQ3: What performance (accuracy) have these methods achieved?

RQ4: What are the main limitations and future directions?

1

Start with WHO, WHAT, HOW, WHY, or WHICH:

Good: 'Which CNN architectures are used for X?' Bad: 'Is deep learning good?'

2

Make it answerable from existing literature:

You should be able to answer it by reading papers — not by running experiments.

3

Keep it specific:

'How does ResNet perform on plant disease datasets compared to VGGNet?' — not 'How is deep learning?'

4

Aim for 2–4 questions:

Too few = not enough to write about. Too many = impossible to answer properly in 7–8 pages.

Step 3: Search the Literature

Where and How to Find Research Papers

Google Scholar

Best starting point. Free. Covers almost everything.

IEEE Xplore

Essential for CSE topics. High-quality technical papers.

Semantic Scholar

AI-powered. Shows citations and related papers.

PubMed

Best for medical/health AI topics.

How to Search Effectively

Use keywords + year filter:

"image classification" AND "deep learning" — then filter: 2019–2024

Use Boolean operators:

AND narrows results. OR broadens them. NOT excludes terms.

Search by title / abstract:

Tick 'Title only' when results are too many and unfocused.

Follow citations:

Find one good paper → look at what IT cites → and who cites IT.

Set a language filter:

Filter to English only unless you read another language.

Step 4: Select Your Papers (5–8 Papers)

How to Choose Which Papers to Include

You may find 50–200 results when you search. You do NOT read all of them. You must filter down to 5–8 high-quality, relevant papers.

INCLUDE a paper if...

- It directly answers one of your research questions
- It is peer-reviewed (published in a journal or conference)
- It is published in the last 5–7 years (prefer recent)
- It is written in English
- It includes methods, results, and conclusions
- It is cited by other papers (has credibility)
- You can access the full PDF

EXCLUDE a paper if...

- It is about a different topic / not directly relevant
- It has no peer review (blog posts, Wikipedia, news)
- It is older than 10 years (unless a key founding paper)
- It is in a language you cannot read
- It only has an abstract — no full paper available
- It has very few citations and unknown source
- It is a duplicate of another paper you already have

Step 5: Read the Papers & Take Notes

Don't Just Read — Extract Information Systematically

Never read a paper without taking structured notes. Create a table (Synthesis Matrix) for every paper you read.

Paper	Authors / Year	Method Used	Dataset	Key Results	Limitation
AlexNet	Krizhevsky 2012	CNN (8 layers)	ImageNet	15.3% Top-5	High memory
ResNet	He et al. 2016	ResNet-50	ImageNet	5.25% Top-5	Black-box
VGGNet	Simonyan 2015	16-layer CNN	ImageNet	7.3% Top-5	138M params

↑ *Create this table in Excel or Word. Fill in one row per paper. This becomes the backbone of your review.*

How to Read a Paper Efficiently (Don't Read Word-for-Word!)

Pass 1 — Skim (5 min):

Read ONLY title, abstract, and conclusion. Decide: Is this paper relevant? YES → continue. NO → discard.

Pass 2 — Structure (10 min):

Read section headings and look at figures/tables. Understand what the paper is doing.

Pass 3 — Deep Read (30 min):

Read the full paper. Take notes in your Synthesis Matrix. Highlight key numbers and findings.

Step 6: Organise Your Papers by Theme

Group — Don't Just List Papers One by One

⚠ The #1 Beginner Mistake: Writing the Literature Review as a list — 'Paper 1 found X. Paper 2 found Y. Paper 3 found Z.' — This is NOT a review. It is an annotated bibliography.

✓ The Right Approach: Group papers by THEME (what they have in common), then discuss each group together.

Example — How to Group 6 Papers on Plant Disease Detection:

Theme 1: CNN-Based Methods

Papers [1][2][3]

Discuss what all three CNN papers have in common: architecture type, dataset used, accuracy. Compare their results.

Theme 2: Transfer Learning Methods

Papers [4][5]

Discuss how these two papers use pretrained models. What did they transfer? How did it help accuracy?

Theme 3: Lightweight/Mobile Models

Paper [6]

Discuss the need for mobile-friendly models. What trade-offs did this paper make between size and accuracy?

Step 7: The Structure of Your Review Paper

Every Section Explained — What Goes Where

1

Title

1 line

Specific, clear, includes topic and method. E.g.: 'A Review of Deep Learning Methods for Plant Disease Detection'

2

Abstract

150–250 words

Mini-summary: background, research questions, how you searched, key findings, conclusion. Write this LAST.

3

Introduction

~1 page

Introduce the topic, why it matters, what gap exists, and state your research questions. End with paper structure.

4

Literature Review

3–4 pages

The main body. Organised by theme. Compare, contrast, and synthesise papers — don't just summarise each one.

5

Discussion

~1 page

What do the findings mean overall? What patterns emerged? What do the papers agree/disagree on? What is missing?

6

Conclusion

~0.5 page

Answer your research questions directly. Summarise what you found. Suggest future research directions.

7

References

1 page

List every paper you cited. Use IEEE format for CSE. Must be consistent and complete. Use Zotero or Mendeley.

Writing the Introduction

How to Write It — With a Real Example

The introduction has 5 parts. Think of it as a funnel: start broad, narrow to your specific topic.

A

Opening / Hook:

Start with a big statement about why this topic matters to the world.

"Plant diseases destroy up to 40% of global food production annually, threatening food security for billions of people."

B

Background Context:

Briefly explain the background: what is the problem and why is it hard?

"Traditional disease detection relies on human visual inspection, which is slow, expensive, and prone to error."

C

Role of Technology:

Introduce how deep learning or AI is being used to solve this problem.

"Recent advances in deep learning, particularly CNNs, have shown promising results in automated visual disease recognition."

D

Research Gap:

Identify what is missing: what has not been reviewed or is unclear?

"However, no comprehensive review exists comparing these methods across diverse crops and environmental conditions."

E

Objective & Questions:

State clearly what YOUR paper will do and list your research questions.

"This paper reviews CNN-based methods for plant disease detection. We address the following research questions: RQ1..."

Writing the Literature Review Section

The Heart of Your Paper — Most Challenging Part

✗ WRONG WAY (Avoid This)

Listing papers one by one:

Krizhevsky et al. (2012) proposed AlexNet which achieved 15.3% top-5 error on ImageNet.

Simonyan and Zisserman (2015) proposed VGGNet with 16 layers and achieved 7.3% error.

He et al. (2016) proposed ResNet and achieved 5.25% error rate using skip connections.

This tells the reader nothing useful. There is no comparison, no theme, no insight.

✓ RIGHT WAY (Synthesis)

Grouping and comparing by theme:

Several studies have demonstrated that increasing network depth significantly improves classification accuracy. AlexNet [1] established a baseline of 15.3% top-5 error on ImageNet. VGGNet [2] later showed that stacking small 3×3 filters to 16 layers reduced this to 7.3%, while ResNet [3] achieved a further improvement to 5.25% by introducing skip connections that resolved the vanishing gradient problem. Notably, ResNet achieves better accuracy than VGGNet despite having far fewer parameters, suggesting that architectural innovation matters more than raw depth.

This groups papers by theme, compares their results, explains why one is better, and uses ALL three citations in one paragraph.

Academic Writing: Language & Style

How to Sound Like a Researcher



Use hedging language

Research papers never claim 100% certainty. Use: 'Studies suggest...', 'Results indicate...', 'Evidence shows...' — not 'It is proven that...'



Use formal language

Avoid: 'don't', 'it's', 'a lot', 'really', 'good'. Use: 'do not', 'it is', 'significantly', 'notably', 'effective'. Never use slang.



Write in third person

Avoid 'I read papers and found...'. Use 'This review examines...', 'The selected studies demonstrate...', 'Analysis reveals...'



Use transition phrases

Connect ideas: 'In contrast...', 'Similarly...', 'Building on this...', 'Despite these results...', 'Furthermore...' — never just start a new paragraph with a new paper name.



Be precise and specific

Say 'achieved 95.2% accuracy on the PlantVillage dataset' — not 'achieved high accuracy'. Every claim should have a number or citation.



One idea per sentence

Long sentences with many commas are hard to follow. Write short, clear sentences. Each sentence should have ONE main idea.



Useful phrase starters: 'This review identifies...' · 'Existing studies reveal...' · 'A common finding across the literature is...' · 'In contrast to [X], [Y] demonstrates...'

Step 8: Citations & References

How to Cite Correctly — IEEE Format for CSE

Every fact, result, or idea taken from another paper MUST be cited. Failure to do so is plagiarism — even if you rewrite the sentence.

IEEE In-Text Citation Format (used in body of paper):

Rule: Use square brackets with a number. The number matches the order you first cite the paper.

Example: AlexNet achieved a top-5 error rate of 15.3% on ImageNet [1]. ResNet later improved this to 5.25% using skip connections [2].

Multiple citations: CNNs have been widely applied to image classification [1],[2],[3].

IEEE Reference List Format (at the end of paper):

Journal Article:	A. Krizhevsky, I. Sutskever, and G. E. Hinton, "ImageNet classification with deep CNNs," Commun. ACM, vol. 60, pp. 84–90, 2017.
Conference Paper:	K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in Proc. IEEE CVPR, Las Vegas, NV, USA, Jun. 2016, pp. 770–778.
Book:	I. Goodfellow, Y. Bengio, and A. Courville, Deep Learning. Cambridge, MA: MIT Press, 2016.

Common Mistakes to Avoid

Every Beginner Makes These — Learn From Them Now

1

✗ Listing papers instead of synthesising

✓ Group papers by theme. Compare and contrast. Ask: What do these papers agree on? Where do they differ?

3

✗ Using non-academic sources

✓ No Wikipedia, blog posts, or news sites. Use only journal articles, conference papers, and books.

5

✗ Including too many or too few papers

✓ Stick to 5–8 papers. Quality over quantity. 5 deeply understood papers beat 15 poorly understood ones.

7

✗ Writing the abstract first

✓ Write the abstract LAST. You cannot summarise what you have not yet written.

2

✗ Copying text from papers (plagiarism)

✓ ALWAYS paraphrase in your own words. Then add the citation. Never copy a sentence — even if you cite it.

4

✗ Writing without structure

✓ Write your outline first. Know what each paragraph will say BEFORE you start writing the paragraph.

6

✗ Ignoring the research questions

✓ Constantly refer back to your RQs. Every paragraph should help answer one of your research questions.

8

✗ No proofreading before submission

✓ Read aloud. Use Grammarly. Ask a classmate to read it. Never submit a first draft.

Step 9: Revise Before You Submit

Your Final Quality Checklist

Before submitting, go through every item below. If ANY box is not checked, fix it first.

Content & Structure

- ☐ Title is specific and reflects the paper content
- ☐ Abstract covers: topic, RQs, search method, key findings, conclusion
- ☐ Introduction has background, gap, and clearly stated research questions
- ☐ Literature review is organised by theme — NOT paper-by-paper
- ☐ Discussion answers each research question explicitly
- ☐ Conclusion summarises findings and suggests future work

Citations & Formatting

- ☐ Every claim from a paper has an in-text citation
- ☐ All in-text citation numbers match the reference list
- ☐ Reference list is in IEEE format and alphabetically/numerically ordered
- ☐ Page numbers, margins, and font size follow guidelines
- ☐ Figures and tables are labelled and cited
- ☐ No plagiarism — all text is in your own words

Language & Polish

- ☐ No contractions (don't → do not, it's → it is)
- ☐ No first-person unless allowed ('I' → 'This review')
- ☐ Transitions between paragraphs and sections are smooth
- ☐ All sentences are clear, precise, and grammatically correct
- ☐ Checked with Grammarly or similar tool
- ☐ Asked a classmate or friend to read and give feedback

Recommended Free Tools

Use These to Make Your Work Easier

Finding Papers

Google Scholar

scholar.google.com — Search any topic. Best starting point.

IEEE Xplore

ieeexplore.ieee.org — Top-quality CSE papers.

Semantic Scholar

semanticscholar.org — See who cited what. Find related papers.

Unpaywall (browser ext.)

Gets free PDF for papers behind paywalls.

Organising References

Zotero (FREE)

Saves papers, generates IEEE citations automatically. Best tool.

Mendeley

Similar to Zotero. Good PDF reader + citation manager.

Excel / Google Sheets

Make your Synthesis Matrix. Track all papers in a table.

Notion / OneNote

Take reading notes. Organise by paper or theme.

Writing & Checking

Microsoft Word

Use IEEE template. Track changes for revision.

Overleaf (LaTeX)

Professional typesetting. Free for students. IEEE template ready.

Grammarly

Free grammar and clarity checker. Use before submitting.

Turnitin / iThenticate

Check plagiarism percentage before your supervisor does.

Sample Writing Schedule

Plan Your Time — Don't Write Everything in One Night

Week 1	Choose topic + form research questions + get approval	3–4 hrs
Week 2	Search literature + build list of 15–20 candidate papers	4–5 hrs
Week 3	Filter to 5–8 papers + read and complete Synthesis Matrix	6–8 hrs
Week 4	Organise by theme + write outline + write Literature Review	5–6 hrs
Week 5	Write Introduction + Discussion + Conclusion	4–5 hrs
Week 6	Format references + write Abstract + check for plagiarism	3–4 hrs
Week 7	Full revision + proofreading + peer review + final submission	2–3 hrs

S U M M A R Y

You Can Write a Review Paper.

Follow the 9 steps. One step at a time.

- ① Choose a specific topic you understand
- ② Form 2–4 clear research questions
- ③ Search IEEE Xplore / Google Scholar
- ④ Select 5–8 quality peer-reviewed papers
- ⑤ Read with a Synthesis Matrix
- ⑥ Organise by theme — synthesise, don't list
- ⑦ Write using formal academic language
- ⑧ Cite every source in IEEE format
- ⑨ Revise, proofread, then submit

Good luck! 🍀 Remember: Every expert was once a beginner who started with step 1.